Ian Abiel Wangsa

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Education

National University of Singapore (NUS) Singapore
BComputing in Computer Science w/ Major in Mathematics & Minor in Economics |GPA: 4.50/5.00 Aug 2025 – Aug 2029

- **Relevant Coursework:** Programming Methodology I & II, Data Structures & Algorithms, Linear Algebra, Discrete Structures, Calculus.

Eunoia Junior College Singapore
GCE A-Levels (H2 Mathematics, Physics, Chemistry, Economics) Jan 2021 – Nov 2022

- **Academic Honors:** Dean's List (2021, 2022); MOE Independent Schools Scholarship (2021); MOE Edusave Merit (2022); MOE Certificate of Academic Achievement (2019, 2021, 2022) & MOE Good Award (2017, 2019, 2021).
- **Varsity Softball:** 1st Runner-Up in National School Games (2022); Recipient of the Singapore Schools Sports Council National Colours Award.

Competitions 2015

- Bronze, National Mathematical Olympiad of Singapore (NMOS); Top Brain of School, KooBits Maths Challenge (Season 1).

Experience

Caerus Global Management Singapore
Quantitative Summer Analyst – Apprenticeship Aug 2026 – Sep 2026

- Built async Python/WebSocket L2 feeds for Binance/Hyperliquid with aiohttp, orjson and Parquet.
- Aligned Python/WebSocket L2 books at 100 ms via snapshot–delta reconciliation, sequence checks and feed recovery.
- Computed microprice, order-flow imbalance (OFI) and cross-venue basis with pandas/NumPy.
- Estimated a 0.7–0.8 s Binance lead over Hyperliquid with Python/pandas lead-lag tests and walk-forward validation.
- Modeled ladder-walking impact, maker/taker fees and latency in Python/NumPy backtests of signal decay, funding breakeven and liquidity sensitivity.

Stealth Startup Singapore
Product Engineer Intern Jul 2026 – Present

- Automated Go-service E2E tests with JavaScript/Playwright, REST assertions and SQL checks.
- Validated 23 authentication, onboarding and authorization flows with Playwright screenshots and SQL assertions.
- Built SQL/Grafana dashboards for WAU/MAU, registration funnels and cohorts with scheduled KPI exports.
- Fixed calendar sync, unread-message state and sign-in bugs in React/Next.js and TypeScript.
- Shipped React Native iOS features with typed REST clients and native auth for an Apple App Store release.

CTES Consulting Singapore
Software Engineer Intern Jul 2026 – Present

- Built React/TypeScript/Vite recruitment portals with Tailwind CSS for candidates, clients and consultants.
- Implemented Express REST APIs, PostgreSQL migrations and Cognito JWT/RBAC across frontend/backend services.
- Built Python/Playwright sourcing with BM25 ranking, pypdf/python-docx parsing and openpyxl exports.
- Cut candidate profiling from 1–2 days to hours with Python/BM25 ranking and React shortlist dashboards.
- Dockerized Node.js/Express and Python profiling services with Docker Compose.
- Automated GitHub Actions CI/CD for TypeScript checks, tests and migrations with AWS OIDC, ECR/EC2 and S3/CloudFront.

Accenture Singapore
Application Engineer Intern, Technology (Health & Public Services) Apr 2026 – Jul 2026

- Built a Python hybrid-RAG agent with BGE/BM25 retrieval for root-cause analysis over code and logs.
- Cut incident testing from days to under 30 minutes with Claude CLI agents and PEGA MCP test generation.
- Maintained Git/Azure DevOps regression and integration workflows for Java/PEGA applications.
- Integrated React/JavaScript/HTML with Java/PEGA through a PEGA MCP server and WebSockets.
- Resolved production defects and maintained Flask/Python/Playwright SharePoint retrieval.

D+A Strategies Remote
Quant Trainee – Apprenticeship Mar 2026 – Present

- Implemented ARIMA, Bollinger-band and EMA-pullback strategies in Python/Backtrader with pandas/NumPy feature pipelines and statsmodels forecasting.
- Computed Sharpe ratios, drawdowns and buy-and-hold benchmarks with Backtrader, pandas and NumPy.
- Evaluated ATR stops and downside risk in Python/Backtrader historical simulations.

Technical Skills

Languages: Python, C++20, Java, Go, TypeScript, JavaScript, SQL, C#, Bash, HTML/CSS.
Frontend: React, Next.js, React Native, Expo, Tailwind CSS, Zustand, TanStack Query.
Backend & APIs: Node.js/Express, FastAPI, Flask, Spring Boot, PEGA, REST, WebSockets, SSE, Pydantic, SQLAlchemy/SQLModel, pybind11.
ML & AI: PyTorch, scikit-learn, LightGBM, XGBoost, NumPy, pandas, SciPy, statsmodels, Transformers, NeMo, YOLO, SB3, RAG/BM25, embeddings, fine-tuning, quantized inference.
Quant & Research: VaR/ES, Monte Carlo, option Greeks, GARCH/HMM/Kalman, walk-forward validation, Backtrader.
Data: PostgreSQL, Supabase, SQLite, Redis, Parquet, Neo4j, ChromaDB.
Cloud & DevOps: Docker Compose, Terraform, Linux, GitHub Actions, Azure DevOps, AWS (EC2, ECR, S3, Cognito, RDS, DynamoDB, CloudFront), OIDC, Cloudflare, Firebase.
Testing & Tools: Git, Playwright, Pytest, Vitest, CMake, Jupyter, Streamlit, Grafana, CI/CD, MCP, Unity/C#, BLE.
Certifications: AWS Cloud Practitioner (CLF-C02), Databricks AI/BI, Cisco Specialist, Anthropic Claude 101, GitHub.

Projects & Hackathons

BrainHack TIL-AI 2026 — DSTA Singapore | *1st Place (Team Pandamonium)* May 2026 – Jun 2026
- **Tech Stack:** Python, Docker, FastAPI, PyTorch, NeMo, YOLOv11, BM25, Transformers, SB3.
- Built a finals-ready multimodal AI system with 5 Dockerised FastAPI microservices deployed in an air-gapped environment.
- **ASR:** Fine-tuned NVIDIA Parakeet TDT with custom decoder, FP16 inference, sorted batching, and phrase correction.
- **CV:** Engineered a YOLOv11 target detection pipeline for aircraft, vehicles, and vessels with threshold tuning.
- **Noise:** Built a YOLO-surrogate PGD adversarial noise generator constrained by SSIM/RMSE budgets to fool opponent detectors.
- **NLP:** Deployed a BM25-first RAG pipeline with alternative neural reranking layers and quantized LLM generation.
- **AE:** Developed a MaskablePPO RL navigation agent featuring action masking, reward shaping, and fallback BFS/A* pathing.

AlphaEngine – Quant Research & Execution Platform Aug – Sep 2026
- **Tech Stack:** Python, C++, TypeScript, Next.js, FastAPI, pybind11, PostgreSQL/Supabase, Docker, WebSockets.
- Streamed Binance/Bybit/Hyperliquid L2 data via Python/WebSockets and aiohttp/orjson to compressed Parquet.
- Reconciled 100 ms Python/WebSocket books with snapshot–delta updates, sequence checks and feed recovery.
- Integrated FastAPI/Pydantic risk APIs, a Next.js/React/TypeScript frontend and pybind11 C++ bindings.
- Validated REST schemas and bit-exact C++/Python numerical parity with shared regression fixtures.
- Deployed a Docker/Supabase PostgreSQL gateway with row-level security (RLS), order logs and pre-trade checks.
- Built Python/Telegram Bot API alerts for orders, positions and P&L telemetry.

Obital – NUS Orbital, Apollo 11 May – Aug 2026
- **Tech Stack:** React, Vite, TypeScript, Tailwind CSS, Zustand, FastAPI, DeepSeek, SQLite/Supabase, SQLAlchemy, AES-GCM.
- Built an offline-first React/Vite/TypeScript study app with Tailwind CSS and Zustand.
- Built FastAPI/Pydantic document, chat and study-progress APIs with SQLAlchemy ORM.
- Parsed and indexed PDF/slide chunks with MiniLM and Web Workers for semantic retrieval in React/TypeScript.
- Secured SQLite/Supabase workspaces with Argon2id key derivation, AES-GCM encryption and optional account-scoped cloud sync.
- Integrated DeepSeek LLMs, SSE streaming and tool calling for grounded tutoring and bounded lesson/quiz generation.
- Wired React/TypeScript grading, XP, streaks and leaderboards to FastAPI with Pytest/CI coverage.

Technical Leadership & Co-Curricular Activities

NUS Student Investment Fund | Quantitative Developer Analyst Aug 2026 – Present
- PM competition: Added Python/ib_insync NAV collectors and SQLite/Streamlit dashboards for IBKR paper portfolios.
- PM risk: Computed historical and 50,000-path Monte Carlo VaR/ES plus exposure/drawdown metrics with Python/NumPy.
- Internal competition: Supported React/Streamlit leaderboards with SPY/QQQ benchmarks and Sharpe/Sortino/Calmar ratios.
- Scheduled SQLite WAL NAV snapshots and portfolio reports with systemd services/timers.
- EC2 recovery: Contributed Terraform Auto Scaling, Docker/IB Gateway and DynamoDB conditional leases with fencing tokens for single-worker failover.
- Stress scenarios: Contributed React/Node.js and Python/IBKR dashboards for 19 crises and eight macro factors, covering option Greeks, payoff bounds and VaR/ES.

NUS Fintech Society | Machine Learning Trainee Aug 2026 – Present
- Trained scikit-learn credit-risk models on six datasets using logistic regression, ensembles, SVMs and MLPs.
- Built scikit-learn Pipeline/ColumnTransformer preprocessing with stratified 60/20/20 splits and cross-validation.
- Achieved 0.952 held-out HMEQ ROC-AUC with scikit-learn gradient boosting, validation-only model/threshold selection and PR-AUC/Brier/calibration checks.
- Built k-means/PCA/GMM/Isolation Forest pipelines in scikit-learn/NumPy with reproducible Jupyter experiments.

NUS MedTech – PneuRelief | Software Engineer Aug 2026 – Present
- Develop React Native/Expo and FastAPI cast-pressure monitoring with TypeScript and Supabase/SQLite.
- Implement BLE GATT notifications and Base64/CSV decoding with TypeScript/react-native-ble-plx.
- Manage device commands, Android permissions and BLE reconnection in React Native/TypeScript.

Google Developer Group @ NUS Computing | Software Engineer Aug 2026 – Present
- Build Caring For Life volunteer workflows with React/TypeScript/Vite and TanStack Query.
- Contribute FastAPI/Pydantic APIs with Firebase Auth and SQLModel/PostgreSQL persistence.
- Validate typed React/TypeScript REST clients with Pytest/Vitest frontend/backend regression tests.

NUS Games Development Group | Software Engineer, Project Pandamonium Jan 2026 – Present
- Built procedural 2D board rendering in Unity/C# for Project Pandamonium.
- Centralized Unity/C# state updates to synchronize board rendering and gameplay.

Additional Leadership

Republic of Singapore Air Force (RSAF) & Singapore Army Singapore
Air Force Signals Commander & Army Specialist Cadet (NSF) *Mar 2023 – Jan 2025*
- **Key Deployments:** Selected for Exercise Wallaby (Rockhampton, Australia) to establish joint SAF–RAAF radio communications; participated in SAF Day Parade (Jul 2024).
- **Unit Leadership:** Served as Comms Store IC, Gym Innovation IC, and National Education IC, leading 160 SQN to win the NE CDF Award for two consecutive years (2023–2025); received an Outstanding Performance and Conduct letter.

NUSSU VENtures | *Operations Executive* Nov 2025 – Oct 2026
- Coordinate event planning, venue arrangements and cross-functional communication among organising teams and stakeholders.

NUS Sports Club – Project RunNUS | *Admin & Finance Executive* Jan 2026 – Sep 2026
- Managed event budgets, financial tracking and vendor procurement for a campus run engaging 1,000+ runners.

NUS Sports Club – Project SunNUS | *Team Sports Series Executive* Jan 2026 – Jun 2026
- Coordinated tournament fixtures, venue setup and event logistics for a beach sports event with 400+ participants.

NUS Community Service Club – Project BHID | *Programmes Head* Aug 2025 – Aug 2026
- Led programme design for elderly beneficiaries with intellectual disabilities and directed student volunteer training sessions.

Youth Corps Singapore & NUS Rotaract Club | *Volunteer Educator* Aug 2025 – Jan 2026
- Taught Scratch programming, computational thinking and foundational AI concepts through primary-school workshops.